Traffic Impact Study Guidelines

Updated: June 2025

Introduction

This document outlines the trip generation study (TGS) and traffic impact study (TIS) requirements for Vineyard City. The purpose of this document is to guide City staff and traffic consultants in knowing how to scope traffic studies for new development within the City. The guidelines in this document were based on Utah Department of Transportation (UDOT) and Institute of Transportation Engineers (ITE) guidelines.

Application Submittal and Review

Prior to submitting a site plan or subdivision application to Vineyard City, the applicant should meet with the City's Development Review Committee where they will provide information about the nature and location of the development. Required information will include but may not be limited to:

- Location of development
- Land use type (e.g. single-family or multi-family housing, office, retail, mixed-use, etc.)
- Land use intensity (e.g. unit count, square footage)
- Proposed access points

Based on the information submitted by the applicant, City staff will decide which level of traffic study is required and the boundaries of the study area. Prior to the traffic study beginning, the applicant should contact City Staff to identify the scope of the traffic study. Additional items required for this study will be identified by Vineyard City staff which could include a request for ADT counts, speed data collection / evaluation, intersection sight distance review, traffic control review, etc.

Trip Generation

A trip represents a vehicle entering or exiting a project site. Trip generation is determined using the ITE *Trip Generation* manual, which contains trip rates for land uses based on actual counts of trips at various study sites around the United States. The latest *Trip Generation* manual should be used to calculate new trips for a development (currently 11th Edition, 2021). After calculating the trip generation, all trip counts should be rounded up to a whole number.

The ITE *Trip Generation* data provides an average trip rate per an independent variable (e.g. units, square feet) that can be used. If there are several data points for a land use, ITE also provides a fitted curve equation, which is either a linear or logarithmic equation that may provide a more accurate estimate of new trips for a development. The fitted curve equation is reported with a coefficient of determination (R^2) value, which is a value between 0 and 1 that indicates how well the equation fits the data points, with "1" representing a great fit. If the data provided by ITE is not good, the traffic engineer may need to collect local trip data.

The following are various considerations that should be made when determining whether to use the fitted curve equation, the average rate, or local trip data for traffic studies (see ITE *Trip Generation* Handbook, Chapter 4 for more details):

Use the fitted curve equation when:

- A matching ITE land use has at least 20 data points
- OR the R^2 value is at least 0.75

Use the average rate when:

- A matching ITE land use has at least three (and preferably six) data points and the fitted curve equation is not appropriate
- AND the standard deviation is less than 55 percent of the average rate value

Use local trip data when:

- The development land use is unique and does not match any ITE land uses
- OR a matching ITE land use has less than three (and preferably six) data points
- OR if neither the average rate line nor fitted curve fall within the data cluster
- OR the size of development is outside the range of land use sizes in the ITE data

In addition to the above considerations, the engineer should also visually observe where a development fits along the trip generation average rate line or fitted curve to see that the line/curve falls within the cluster of data points of that particular development size. For example, if the fitted curve is deemed appropriate based on data points or the R^2 value but the average rate line fits the data cluster better for that development size, the average rate may be a better option, or vice versa.

Certain developments may experience a reduced new trip generation number due to the nature of the project. The following are the trip reductions that should be considered in trip generation and traffic impact studies based on standard ITE methodologies:

- Pass-by trips: Trips made to a development on the way to an existing destination
- Multi-modal: Trips made via non-vehicle modes such as transit, walking, and biking
- Mixed-use internal capture: Trips made within the development itself due to a mix of uses

For commercial and office space, the City requires that square footage be used as the independent variable to calculate trips, as opposed to using number of employees or other variables. This will help ensure a conservative calculation of trips for a development that will be valid even if the tenant changes in the future.

If an applicant believes they will generate less trips than the average similar land use, a professional traffic operations engineer (PTOE) can complete a local trip generation study based on trip counts and square footages of at least two similar sites along the Wasatch Front.

Traffic Study Levels

Small developments may not need a full TIS but may still be required to submit a TGS prior to City approval. A TGS includes a description of the project and a calculation of the anticipated trip generation. In addition to the elements of a TGS, a TIS includes level of service (LOS) analyses for study intersections for the current year and potentially future years and recommendations to mitigate poor levels of service. For Vineyard City, the acceptable LOS threshold varies by road functional classification in location in the City, as outlined in Table 1.

Table 1: Vineyard City Acceptable LOS Thresholds

Location	Local Road	Collector	Arterial
Downtown	C	E	E
Other City Area	B	C	D

A development that is anticipated to generate at least 25 new peak hour trips requires a TIS. A development that generates less than 25 new peak hour trips only requires a TGS. General thresholds of 25 peak hour trips for various common land uses are provided in Table 2.

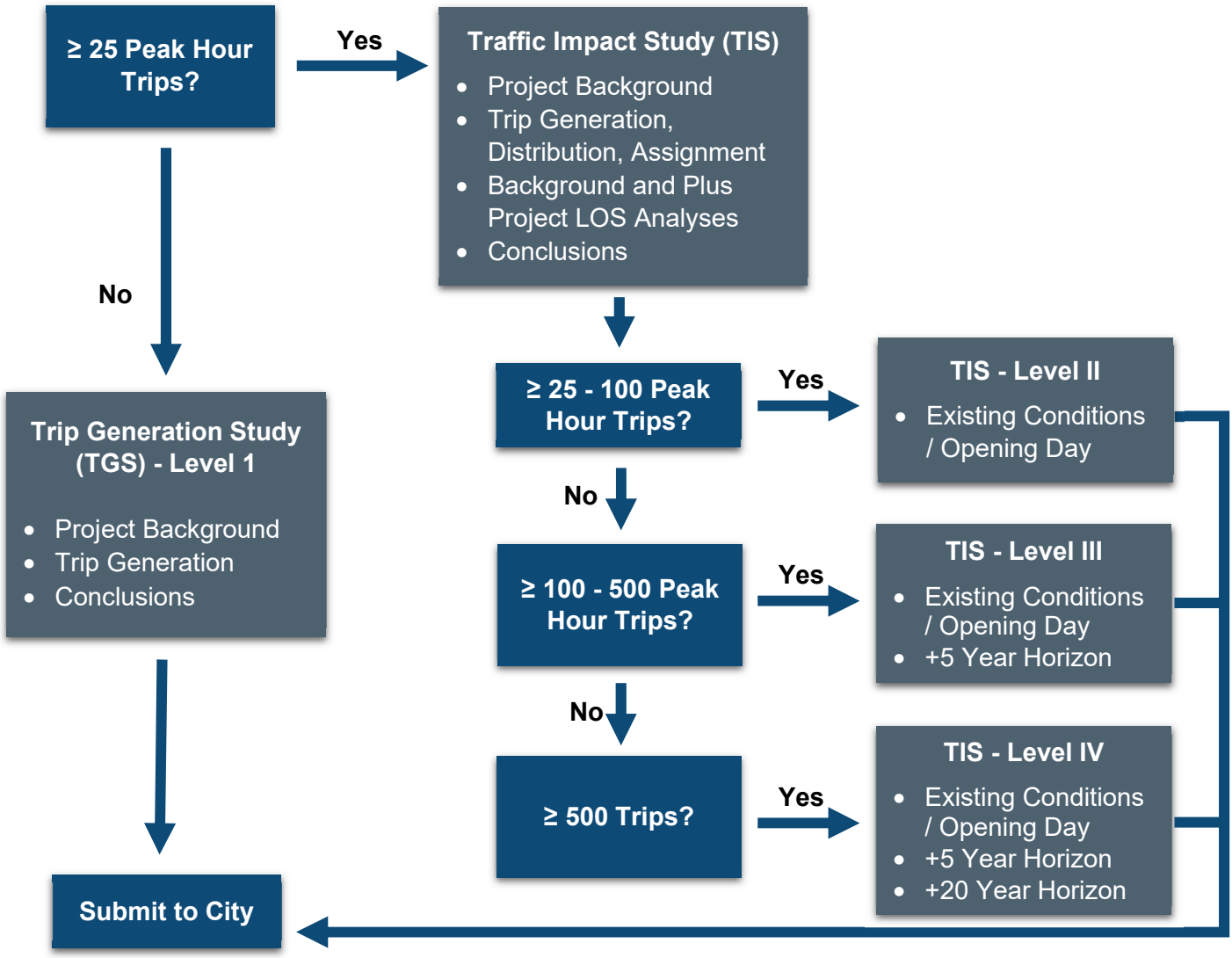
Table 2: General 25 Peak Hour Trip Thresholds

Land Use	ITE Code(s)	≥ 25 Peak Hour Trips
Light Industrial / Manufacturing	110, 140	50,000 sq. ft. GFA
Warehousing	150	150,000 sq. ft. GFA
Mini-Warehouse (Self-Storage)	151	150 storage units
Single-Family Detached Housing	210	25 dwelling units
Single-Family Attached Housing (Townhomes)	215	40 dwelling units
Multifamily: Condo/Apartment	220	50 dwelling units
Mobile Home Park	240	50 dwelling units
Assisted Living	254	100 beds
Hotel	310	*Any Size*
General Office	710	15,000 sq. ft. GFA
Medical / Dentist Office	720	7,000 sq. ft. GFA
Shopping Center	820	5,000 sq. ft. GLA
Bank	911 - 912	*Any Size*
Restaurant	930 - 935	*Any Size*
Gas Station / Convenience Store	944 - 945	*Any Size*

If a TIS is required, City staff will select the level of TIS to be completed based on the anticipated trip generation. At the smallest TIS level (level II), only an analysis of the current or opening year with and without the project is required. At the highest TIS level (level IV), future analyses up to 20 years out with and without the project are also required. A summary of each traffic study level, with its associated peak hour trip thresholds and horizon years, is shown in Table 3. A flowchart to help determine the traffic study level is also shown in Figure 1.

Table 3: Study Horizon Years by Level

Study Level		New Peak Hour Trips	TIS Horizon Year(s)
TGS	I	< 25	N/A
TIS	II	≥ 25 - 100	Current / Opening Year
	III	≥ 100 - 500	Current / Opening Year, + 5 years
	IV	≥ 500	Current / Opening Year, + 5 years, + 20 years



Study Area

The study area to be analyzed in a TIS also depends on the size of the development. Large developments may impact intersections miles away while small developments may have minimal impacts on adjacent roadways. The **minimum** required study areas for each TIS level are provided in Table 4. City staff may request additional intersections as needed.

In general, each TIS will include an analysis of the project accesses and nearby major intersections. Major intersections are defined as signalized or unsignalized intersections of two public roadways that are anticipated to be used by project traffic. This does not include minor intersections or other accesses or private roadways near the project, unless requested by the Vineyard City. However, if a minor intersection is an existing access to the site or located across from an existing or proposed project access, it should be included in the analysis.

Table 4: TIS Study Areas by Level

Study Level	Study Area
II	Project Accesses & Closest Major Intersections
III	Project Accesses & Major Intersections within ¼ Mile
IV	Project Accesses & Major Intersections within ½ Mile

Report Elements

The following sections outline the elements that should be included in the TGS and TIS reports:

TGS Elements

Each TGS report will be required to contain at minimum the following elements:

- Project Background:
  - Identify the project site location, development type, and site access drives
- Trip Generation:
  - Calculate trip generation for the development according to ITE standards
- Conclusions:
  - State the key findings and recommendations of the TGS in a concise manner

TIS Elements

Each TIS report will be required to contain at minimum the following elements:

- Traffic studies should be completed by a licensed professional engineer (PE) with a Professional Traffic Operation Engineer (PTOE) certificate
- The study should be stamp and signed by the PE / PTOE engineer
- Executive Summary:
  - Include a concise summary at the beginning of the report with the key assumptions, findings, and recommendations of the TIS

- Study Area:
  - Identify the existing nearby roadways and intersections including roadway classifications, intersection control, and speed limits
- Data Collection:
  - Collect at minimum peak hour turning movement counts at the study intersections and potentially daily volumes if requested by staff
  - Normally collect morning (7-9 AM) and evening (4-6 PM) peak hour counts unless the study area and/or development will have different peak hours (e.g. schools)
- Analysis Period:
  - Identify the study peak hour based on existing counts and the proposed trip generation; at minimum, the highest peak hour should be analyzed
  - City staff may request that multiple peak hours be analyzed
- Project Background:
  - Identify project site location, development type, project phasing, site access drives and nearby intersections to be affected by the development
- Access and Auxiliary Lanes:
  - Identify the location and configuration of each access drive to the site and the need to add auxiliary lanes (deceleration and acceleration turn lanes)
- Trip Generation:
  - Calculate trip generation according to ITE standards and apply reductions when applicable
- Trip Distribution and Assignment:
  - Document the distribution and assignment of project trips for each horizon year
- Capacity Analysis:
  - Calculate delay, LOS, and queuing results for all study intersections in the study area for existing background (without project) conditions, existing plus project conditions, future background conditions, and future plus project conditions (if required by the study level)
  - Complete this analysis using Synchro/SimTraffic, Vistro, or VISSIM software
  - Report LOS and 95th percentile queueing at study intersections per HCM criteria
- Proposed Mitigations:
  - Identify proposed mitigations to roadway and intersection characteristics in the study area based on the LOS and queueing results
  - Clearly show whether these mitigations are caused by background or site-generated traffic
- Exhibits:
  - Include figures showing peak hour turning movement volumes used in the analysis for each scenario, including trip assignment volumes
  - Include figures or reports that show the assumed lane configurations
  - Include tables or figures that show the LOS at each intersection for each scenario
- Appendices:
  - Include raw traffic count data
  - Include capacity analysis and queueing reports for each scenario
  - Include a site or concept plan of the development if available